

# CUSTOMER SUPPORT TICKET PRIORITY PREDICTION AND AUTOMATED ASSIGNMENT SYSTEM

## SALESFORCE IMPLEMENTATION WITH AGENTFORCE

### Project Documentation

| Project      | CUSTOMER SUPPORT TICKET PRIORITY PREDICTION AND<br>AUTOMATED ASSIGNMENT SYSTEM |
|--------------|--------------------------------------------------------------------------------|
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### SALESFORCE CRM IMPLEMENTATION PROJECT

This document follows the structure and documentation style of the supplied EventForce Management System sample, adapted to the Customer Support Ticket Priority Prediction project.

# USER STORY

The Customer Support Ticket Priority Prediction and Automated Assignment System is designed to provide a centralized Salesforce-based solution for handling customer support tickets. It uses a Support Ticket Intelligence custom object, an auto-launched Flow and Agentforce configuration to retrieve customer/ticket data, analyze issue descriptions, classify priority, and support automated assignment.

## 1. INTRODUCTION

### 1.1 Project Overview

The Customer Support Ticket Priority Prediction and Automated Assignment System is a Salesforce-based solution designed to streamline customer support operations. The system uses a custom Support Ticket Intelligence object, an auto-launched Flow, decision logic, and Salesforce Agentforce to retrieve customer and ticket information, analyze the ticket description, classify the ticket as High, Medium, or Low priority, and route the ticket towards the appropriate support handling process.

### 1.2 Purpose

- Centralize customer support ticket tracking and management in Salesforce.
- Use Agentforce and Flow automation to assess ticket urgency and priority.
- Automatically route high-priority issues to appropriate or senior support handling.
- Reduce manual ticket triage and improve response-time management.
- Maintain structured customer, ticket, priority, status, and assignment information.
- Provide a foundation for reports and dashboards for operational monitoring.

## 2. IDEATION PHASE

### 2.1 Problem Statement

| PS ID | User Role                  | Problem / Need                                                                  |
|-------|----------------------------|---------------------------------------------------------------------------------|
| PS-1  | Support Operations Manager | Support tickets require centralized tracking and consistent prioritization.     |
| PS-2  | Support Agent              | Manual identification of urgent tickets can delay resolution.                   |
| PS-3  | Customer                   | Urgent issues may require faster acknowledgement and handling.                  |
| PS-4  | Support Manager            | Manual ticket assignment can create delays and workload bottlenecks.            |
| PS-5  | System Administrator       | Ticket descriptions need structured analysis through Agentforce and automation. |
| PS-6  | Support Agent              | Priority and SLA-risk information should be visible and trackable.              |

### 2.2 Empathy Map Canvas – Support Agent

Think & Feel: Wants urgent tickets to be identified quickly and handled appropriately.

Hear: Customer complaints, escalations, support updates and priority-related notifications.

See: Ticket details, descriptions, priority, status and assignment information.

Say & Do: Reviews ticket queues, investigates issues, communicates with customers and updates status.

Pain: Manual sorting, delayed identification of urgent issues and workload bottlenecks.

Gain: Automated classification, faster routing and structured support workflows.

### 2.3 Brainstorming & Idea Prioritization

| Approach                     | Key Limitation                                                                  |
|------------------------------|---------------------------------------------------------------------------------|
| Manual Email / Manual Triage | Slow and dependent on human sorting.                                            |
| Basic Assignment Rules       | Does not evaluate the actual ticket description or urgency.                     |
| Standalone AI Tool           | Adds integration and data-governance complexity.                                |
| Salesforce + Agentforce      | Combines CRM data, conversational AI and Salesforce automation in one platform. |

### 3. REQUIREMENT ANALYSIS PHASE

#### 3.1 Customer Journey Map

| Stage | Customer Journey Stage  | What Happens                                                              |
|-------|-------------------------|---------------------------------------------------------------------------|
| 1     | Ticket Submission       | Customer provides account/ticket information and issue details.           |
| 2     | Data Capture            | Account and support-ticket records are stored in Salesforce.              |
| 3     | AI/Logic Analysis       | Agentforce invokes the configured action and Flow to process the request. |
| 4     | Priority Classification | Decision logic classifies the ticket as High, Medium, or Low.             |
| 5     | Automated Routing       | High-priority handling is routed toward senior/appropriate support.       |
| 6     | Agent Engagement        | Support staff investigate the issue and interact with the customer.       |
| 7     | Monitoring              | Ticket status and operational information can be monitored in Salesforce. |
| 8     | Closure                 | The ticket is updated to the appropriate final status after resolution.   |

#### 3.2 Functional Requirements

| FR    | Requirement                                                                                |
|-------|--------------------------------------------------------------------------------------------|
| FR-1  | Support ticket creation and management.                                                    |
| FR-2  | Account/Customer and Contact relationship linking.                                         |
| FR-3  | Automated priority classification using Flow decision logic and Agentforce.                |
| FR-4  | Automated handling/assignment for high-priority tickets.                                   |
| FR-5  | Description analysis using configured conditions such as urgent, not working, and failure. |
| FR-6  | SLA breach risk identification and tracking.                                               |
| FR-7  | Status management using New, In Progress, and Resolved.                                    |
| FR-8  | Agentforce topic and action configuration.                                                 |
| FR-9  | Conversational interaction for account/ticket evaluation.                                  |
| FR-10 | Reports and dashboards for operational visibility.                                         |

#### 3.3 NON-FUNCTIONAL REQUIREMENTS

| NFR             | Description                                                                                          |
|-----------------|------------------------------------------------------------------------------------------------------|
| Usability       | Lightning and Agentforce interfaces should support simple ticket interaction.                        |
| Security        | Customer and ticket data should be protected through Salesforce access controls.                     |
| Reliability     | Auto-launched Flow and decision logic should execute consistently.                                   |
| Performance     | Ticket information should be retrieved and evaluated within normal Salesforce response expectations. |
| Availability    | Salesforce cloud access supports authorized support operations.                                      |
| Scalability     | The data model and automation can accommodate growing ticket volumes.                                |
| Maintainability | Flow logic, Agentforce actions, fields, and rules can be updated as requirements change.             |

#### 3.4 DATA FLOW

Customer / Agentforce → Support Ticket Priority Analysis Topic → Agentforce Action → Auto-Launched Flow → Get Account Record → Get Support Ticket Record → Analyze Description → High / Medium / Low → Assignment / Handling → Final Output → Customer.

Supporting Flow Logic: Account name is received as input; the Flow retrieves the Account, uses the Account relationship to locate support-ticket data, evaluates the description through a Decision element, and returns the configured output to Agentforce.

#### 3.5 Technology Stack

| Layer           | Technology                                                         |
|-----------------|--------------------------------------------------------------------|
| Platform        | Salesforce CRM / Developer Edition                                 |
| AI & Automation | Salesforce Agentforce, Agent Builder, Auto-Launched Flow           |
| UI              | Salesforce Lightning Experience, Agentforce conversation interface |
| Database        | Support Ticket Intelligence__c custom object                       |
| Business Logic  | Get Records, Variables, Decision, Assignment and record operations |
| Security        | Profiles, Roles, Permission Sets and Sharing configuration         |
| Reporting       | Salesforce Reports & Dashboards                                    |

## 4. PROJECT DESIGN PHASE

### 4.1 Problem–Solution Fit

| Problem                             | Solution                                                                     | Benefit                                      |
|-------------------------------------|------------------------------------------------------------------------------|----------------------------------------------|
| Support information is fragmented   | Support Ticket Intelligence custom object with Account/Contact relationships | Centralized ticket information               |
| Manual ticket retrieval             | Agentforce action + Auto-Launched Flow                                       | Faster information retrieval                 |
| Priority identification is manual   | Analyze Description Decision logic                                           | Consistent High/Medium/Low classification    |
| Urgent tickets need faster handling | High-priority assignment path                                                | Supports senior/appropriate support handling |
| Customer interaction is manual      | Agentforce conversational interface                                          | Prompt-based ticket evaluation               |
| Operational visibility is needed    | Salesforce Reports & Dashboards                                              | Structured monitoring                        |
| Access must be controlled           | Salesforce security configuration                                            | Controlled data access                       |

### 4.2 Proposed Solution

| Parameter      | Description                                                                                    |
|----------------|------------------------------------------------------------------------------------------------|
| Problem        | Manual and fragmented support-ticket triage.                                                   |
| Solution       | Salesforce CRM with Agentforce and Flow automation.                                            |
| Innovation     | Conversational ticket retrieval, priority decision logic and automated assignment handling.    |
| Impact         | Reduced manual triage and faster attention to urgent tickets.                                  |
| Business Model | Centralized customer-support ticket management.                                                |
| Scalability    | Custom object and automation can be extended with additional rules, agents and ticket volumes. |

### 4.3 SOLUTION ARCHITECTURE

- Data Layer: Account, Contact and Support\_Ticket\_Intelligence\_\_c.
- Automation Layer: Variables, Get Records, Decision, Assignment and record operations.
- AI / Agent Layer: Agentforce topic, instructions and action linked to the Flow.
- Interaction Layer: Agentforce conversational interface within Salesforce.
- Security & Reporting: Profiles, Roles, Permission Sets, Sharing configuration, Reports and Dashboards.

### 4.4 DATA MODEL

| Field Label     | Data Type        | Purpose                          | Example / Values              |
|-----------------|------------------|----------------------------------|-------------------------------|
| Ticket Number   | Auto Number      | Unique support ticket identifier | TKT-{0000}                    |
| Customer        | Lookup → Account | Related customer/account         | Account record                |
| Contact         | Lookup → Contact | Customer contact                 | Contact record                |
| Issue Type      | Picklist         | Type of support issue            | Technical / Billing / General |
| Description     | Long Text Area   | Detailed issue description       | Free text                     |
| Priority Level  | Picklist         | Predicted/assigned priority      | High / Medium / Low           |
| Status          | Picklist         | Current ticket state             | New / In Progress / Resolved  |
| Assigned To     | Lookup → User    | Support agent assignment         | User record                   |
| SLA Breach Risk | Checkbox         | Indicates possible SLA risk      | True / False                  |
| Created Date    | Date             | Ticket creation date             | Date value                    |

## 5. PROJECT PLANNING & SCHEDULING

### 5.1 Agile Sprint Plan

The implementation is organized as sprint-based work covering data schema, Flow development, decision logic, Agentforce configuration, testing, security and reporting.

| Sprint   | Epic / Module         | User Story / Tasks                                              | Priority |
|----------|-----------------------|-----------------------------------------------------------------|----------|
| Sprint-1 | Data Schema Setup     | Create custom object, fields and Account/Contact relationships. | High     |
| Sprint-1 | Flow Foundation       | Create input/output variables and Get Records logic.            | High     |
| Sprint-2 | Priority Logic        | Implement Analyze Description decision outcomes.                | High     |
| Sprint-2 | Assignment Automation | Configure high-priority handling and support-agent assignment.  | High     |
| Sprint-3 | Agentforce Setup      | Create topic, instructions and action linked to the Flow.       | High     |
| Sprint-3 | Conversation Testing  | Enable output visibility and test Agentforce requests.          | High     |
| Sprint-4 | Security & Reporting  | Configure access controls and operational reports/dashboards.   | Medium   |

## 6. PROJECT DEVELOPMENT PHASE

### Phase 1: Requirement Analysis & Planning

The project scope was defined around centralized support-ticket data, priority classification, automated handling and Agentforce-based conversational access. The implementation follows the supplied project workflow and the EventForce sample's phase-based documentation style.

### Phase 2: Backend Development & Configuration

Milestone 1 – Salesforce Environment: Implemented in a Salesforce environment using the required CRM, Agentforce and Flow configuration.

Milestone 2 – Custom Object: Created Support Ticket Intelligence (Support\_Ticket\_Intelligence\_\_c).

Milestone 3 – Relationships: Customer is linked to Account and Contact is linked through a lookup relationship; Assigned To can reference Salesforce User.

Milestone 4 – Fields: Ticket Number uses Auto Number; Issue Type and Priority Level use Picklists; Description uses Long Text Area; SLA Breach Risk uses Checkbox.

Milestone 5 – Auto-Launched Flow: Uses input/output variables and Get Records to retrieve required Account and ticket information.

Milestone 6 – Decision Logic: Analyze Description classifies High, Medium and Low priority using configured conditions such as urgent, not working and failure.

Milestone 7 – Assignment: High-priority handling supports assignment to the appropriate support agent, including senior support handling.

## 6. PROJECT DEVELOPMENT PHASE – AGENTFORCE CONFIGURATION

### Phase 3: Agentforce Configuration

Milestone 8 – Agentforce Enablement: Agentforce is configured and opened in Agent Builder.

Milestone 9 – Topic: Support Ticket Priority Analysis.

Milestone 10 – Instructions: Mandatory instructions define the topic's responsibility and expected workflow.

Milestone 11 – Action: The Agentforce action is linked to the auto-launched Flow and accepts input such as account name.

Milestone 12 – Conversation Output: Output variables are configured for conversational visibility.

### Phase 4: Execution Workflow

- User opens the Agentforce conversation.
- User requests support-ticket information or priority analysis.
- Agentforce collects the required account input.
- The Agentforce action calls the auto-launched Flow.
- The Flow retrieves the Account record.
- The Flow retrieves the related Support Ticket Intelligence record.
- Analyze Description evaluates the ticket description.
- The Flow follows High, Medium or Low priority logic.
- The appropriate support assignment/handling path is executed.
- The final output is returned to Agentforce.

#### Example – High Priority Ticket

Account Name → Get Account → Get Support Ticket → Analyze Description → High Priority → Senior/Appropriate Support Assignment → Final Response

#### Example – Medium Priority Ticket

Account Name → Get Account → Get Support Ticket → Analyze Description → Medium Priority → Normal Support Handling → Final Response

#### Example – Low Priority Ticket

Account Name → Get Account → Get Support Ticket → Analyze Description → Low Priority → Normal/Lower-Urgency Handling → Final Response

### Phase 5: UI, Security & Reporting

The conversational interaction is exposed through Salesforce/Agentforce. Access can be controlled through Salesforce security settings. Reports and dashboards can be used to monitor ticket volumes, priorities, statuses and assignment information.

## 7. FUNCTIONAL & PERFORMANCE TESTING

Testing focuses on data accuracy, Flow execution, priority decision logic, Agentforce integration, output visibility and access configuration. The supplied implementation transcript demonstrates the end-to-end interaction from account input through ticket retrieval, decision processing and Agentforce response.

| Test Area              | Expected Validation                                                  | Status     |
|------------------------|----------------------------------------------------------------------|------------|
| Object Data Entry      | Required fields and picklists accept valid ticket information.       | Validated  |
| Account Retrieval      | Flow retrieves the account using the supplied account name/input.    | Validated  |
| Ticket Retrieval       | Flow retrieves the related Support Ticket Intelligence record.       | Validated  |
| Priority Decision      | Description conditions route the ticket to High/Medium/Low outcomes. | Validated  |
| High-Priority Handling | High-priority path supports senior/appropriate agent assignment.     | Validated  |
| Agentforce Action      | Conversation input invokes the configured action and returns output. | Validated  |
| Output Visibility      | Configured output variables can be shown in the conversation.        | Validated  |
| Security               | Salesforce access configuration can restrict record visibility.      | Configured |

## 8. ADVANTAGES & LIMITATIONS

### Advantages

- **Automated Triage:** Ticket descriptions are evaluated through configured decision logic.
- **Faster Urgent Handling:** High-priority tickets can follow a dedicated support-agent assignment path.
- **Centralized Platform:** Customer, account and ticket information is maintained within Salesforce.
- **Conversational Access:** Agentforce allows users to request ticket information through natural-language interaction.
- **Scalable Automation:** Additional decision conditions, fields, topics and actions can be added.
- **Operational Visibility:** Reports and dashboards can provide structured monitoring.

### Limitations

- **Developer Edition Scope:** Intended for academic implementation/testing rather than production deployment.
- **Rule Dependence:** Priority quality depends on configured description conditions/keywords and ticket text.
- **Salesforce Configuration Dependency:** Flow, Agentforce, permissions and object configuration require Salesforce knowledge.
- **Production Deployment:** Real-world deployment requires environment-specific security, testing, deployment and governance.

## 9. CONCLUSION

The Customer Support Ticket Priority Prediction and Automated Assignment System demonstrates how Salesforce Agentforce and Flow automation can be combined to streamline customer-support ticket handling. The solution centralizes ticket information in a custom Support Ticket Intelligence object, retrieves related Account and ticket data through an auto-launched Flow, evaluates descriptions through an Analyze Description decision, classifies tickets as High, Medium or Low priority, and supports automated assignment/handling.

The project also demonstrates a conversational Agentforce workflow in which the user provides account information, the agent invokes the configured action, the Flow processes the ticket data, and the final result is returned to the conversation. The architecture provides a foundation that can be extended with richer AI evaluation, additional support rules, dashboards and production-grade security.

## 10. FUTURE SCOPE

- Production Deployment – Move from development/academic environment to governed production architecture.
- Advanced Priority Intelligence – Extend beyond predefined keywords with richer AI-based classification.
- Expanded Routing – Route by agent skills, workload, issue type, language or service category.
- SLA Automation – Add timers, escalation paths and SLA notifications.
- Advanced Analytics – Add dashboards for priority distribution, response time, resolution time and agent workload.
- External Integrations – Connect email, web forms or other support channels.
- Continuous Optimization – Refine decision rules, prompts, topics and Flow logic using operational feedback.

## 11. APPENDIX

### A. Custom Object

- Support Ticket Intelligence (Support\_Ticket\_Intelligence\_\_c)

### B. Core Fields

- Ticket Number
- Customer
- Contact
- Issue Type
- Description
- Priority Level
- Status
- Assigned To
- SLA Breach Risk
- Created Date

### C. Automation

- Auto-Launched Flow
- Variables
- Get Records
- Decision – Analyze Description
- Assignment / Record Operations

### D. Agentforce

- Agentforce Builder
- Support Ticket Priority Analysis Topic
- Mandatory Instructions

- Flow-based Action
- Conversation Output

## **E. Priority Outcomes**

- High Priority
- Medium Priority
- Low Priority

## **F. Security & Reporting**

- Profiles
- Roles
- Permission Sets
- Sharing Configuration
- Reports
- Dashboards

## **G. Testing & Documentation**

- Configuration Testing
- Flow Testing
- Priority Decision Testing
- Agentforce Conversation Testing
- Security Testing
- Project Documentation

**THANK YOU**